LECTURE 20

FINAL PROJECT:

CARETRACE - BUILDING A TRUSTWORTHY PEDIATRIC CARE AGENT, WITH A NEUROSYMBOLIC APPROACH

DATASCI290: NEUROSYMBOLIC AI, SPRING 2026

MARCH 18 2026

THE PROBLEM

- **Turn 1 (11:00 PM)**

  Parent: *My 6-year-old has a fever, threw up once, and looks really wiped out. I'm not sure what to do tonight.*

  CareTrace Agent:

  I can help you decide what's safe to do tonight. I need to quickly check a few important things.

  • What is the temperature right now?

  • Is your child drinking fluids?

  • Are they awake and responding normally when you talk to them?

  • Any trouble breathing or very fast breathing?

  • Is your child currently taking any medications regularly?

THE PROBLEM

- **Turn 2 (11:03 PM)**

  Parent: *Temp is 101.8. He’s tired but answers me. No breathing issues. He’s sipping water, not much though. He’s been on medication for a recent ear infection.*

  CareTrace Agent:

  Thanks , that helps.

  Right now:

  • Fever is moderate

  • He is responsive

  • No breathing problems

  • Drinking some fluids

  I need one clarification about medications:

  • What medication is he currently taking?

  • When was the last dose?

  Also:

  • Has he vomited more than once?

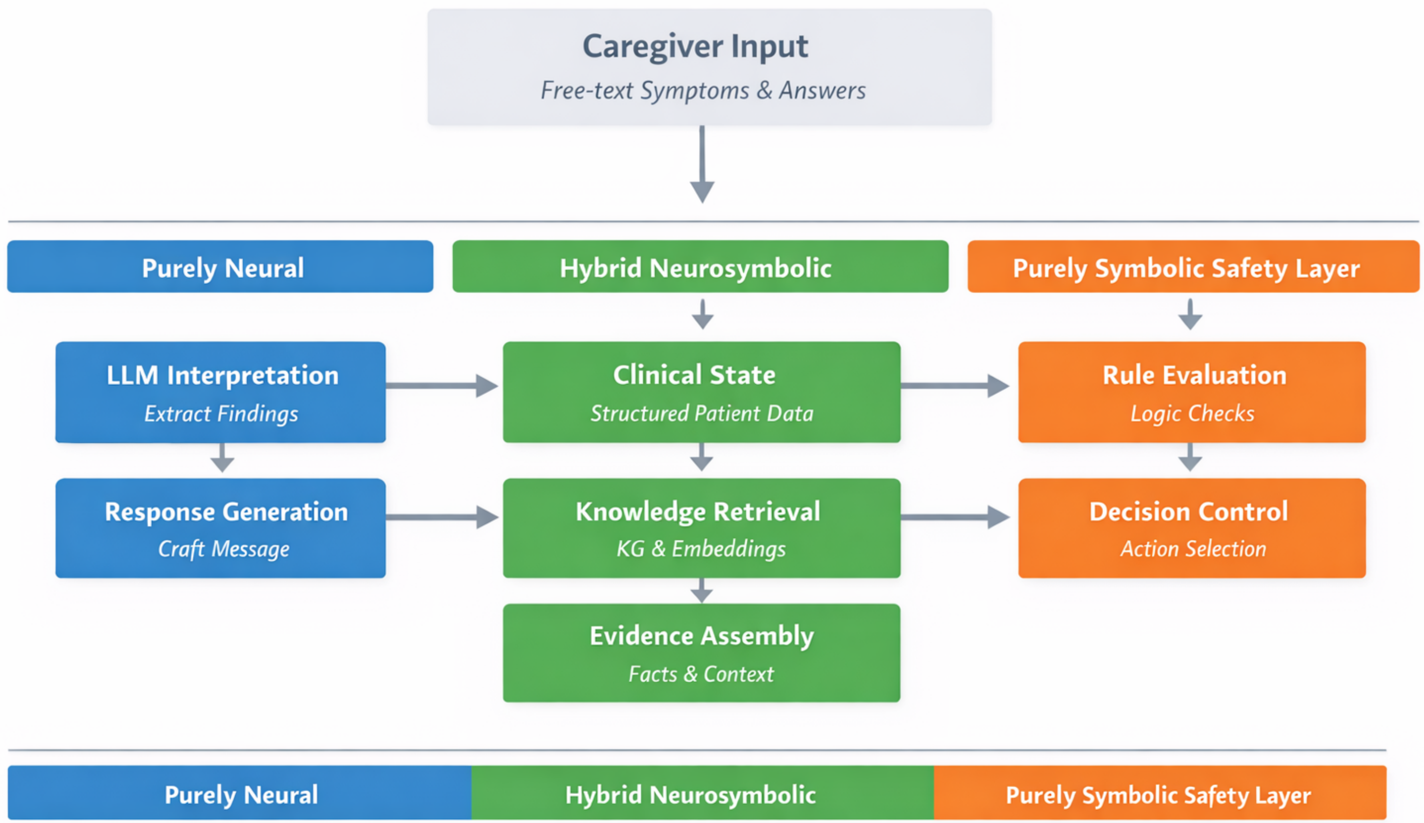
  • Has he urinated in the last 6–8 hours?

THE PROBLEM

- **Turn 3 (11:06 PM)**

  Parent: *He's on amoxicillin. Last dose was earlier tonight.*
  Just vomited once. He peed earlier this evening.
  CareTrace Agent:
  Based on what you've told me, this can be managed at home tonight, but we need to be careful with medications.
  What to do now:
  • Encourage fluids frequently
  • Let him rest
  • Keep clothing light
  For fever discomfort:
  *• Use **only one fever medicine** and follow correct dosing*
  *• If you plan to give something, check that it is **safe with his current medication***
  Watch closely overnight. Seek urgent care or ER if:
  • He becomes hard to wake
  • Breathing becomes fast or difficult
  • Vomiting continues repeatedly
  • He stops drinking fluids or does not urinate for 8 hours
  • Fever rises above 104°F
  If none of these happen, monitoring at home is reasonable.

MULTIPLE COMPONENTS



THE KNOWLEDGE GRAPH

- **Symptoms / findings**
  - Fever: 386661006
  - Lethargy :214264003
  - Dyspnea: 267036007
  - Decreased urine output: 718403007
  - Oliguria: 83128009
  - Drowsiness: 79519003
  - Vomiting:
- **Diagnoses / disorders**
  - Dehydration: 34095006
  - Gastroenteritis: 25374005
  - Viral disease: 34014006
  - Pneumonia: 233604007
  - Community acquired pneumonia: 385093006
  - Bacterial meningitis: 95883001
  - Febrile convulsion: 41497008
- **Medications**
  - Acetaminophen:387517004
  - Ibuprofen:387207008
  - Aspirin:387458008

THE KNOWLEDGE GRAPH

- **Action / Disposition:** These are the final triage decisions or immediate next actions the system can recommend.
  - Examples:
    - Home monitoring tonight
    - Encourage oral fluids
    - Give acetaminophen
    - Give ibuprofen
    - Call pediatrician / doctor
    - Urgent same-day evaluation
- **Supportive Treatment / Home-Care Step:** These are concrete care steps a caregiver can follow at home.
  - Examples:
    - Offer small frequent sips of water / fluids
    - Let child rest
    - Dress child lightly
    - Avoid bundling
    - Use correct dose measuring device
    - Monitor urine output
    - Monitor temperature overnight
- **Triage Questions:** These are the focused follow-up questions the agent can ask.
  - Examples:
    - What is the temperature right now?
    - Is your child drinking fluids?
    - Is your child awake and responding normally?
    - Any trouble breathing or breathing fast?
    - Has your child urinated in the last 8 hours?
    - Has vomiting happened more than once?
    - Any rash?
    - Any pain in the neck, ears, or throat?
- **Decision Criterion / Rule Condition:** These are reusable structured conditions that appear inside rules.
  - Examples:
    - Fever above 104 F
    - Vomiting repeatedly
    - No urine in 8 hours
    - Not alert when awake
    - Trouble breathing
    - Age under 3 months
    - Fever more than 3 days
    - Poor oral intake

THE KNOWLEDGE GRAPH

- **Escalation Level:** This is the severity/disposition tier.
  - Examples:
    - Home care
    - Call doctor
    - Urgent evaluation
    - ER now
- **Monitoring Target:** These are the things the caregiver is told to watch for after the recommendation.
  - Examples:
    - Fluid intake
    - Urination
    - Responsiveness
    - Breathing
    - Temperature trend
    - Repeat vomiting
    - Rash appearance
- **Escalation Trigger:** These are the concrete events or findings that should cause the caregiver to move from home care to urgent care or ER.
  - Examples:
    - Child becomes hard to wake
    - Breathing becomes fast or difficult
    - No urination for 8 hours
    - Vomiting becomes frequent
    - Fever rises above 104 F
    - Rash appears
    - Seizure occurs

STANDARDIZE THE TERMINOLOGY

- *“A fever is a body temperature of over 100.4 degrees F”*
  - CPG phrase
    - “fever”
    - “body temperature of over 100.4 degrees F”
  - KG concepts this maps to
    - SNOMED Finding: Fever
    - Manual Decision Criterion: Temperature > 100.4 F
    - Manual Triage Question: “What is the temperature right now?”

STANDARDIZE THE TERMINOLOGY

- *"You can give acetaminophen… You can give acetaminophen or ibuprofen if your child is over 6 months old"*
- CPG phrase
  - "acetaminophen"
  - "ibuprofen"
  - "over 3 months"
  - "over 6 months"
- **KG concepts this maps to**
  - **SNOMED Medication:** Acetaminophen
  - **SNOMED Medication:** Ibuprofen
  - **Manual Decision Criterion:** Age > 3 months
  - **Manual Decision Criterion:** Age > 6 months
  - **Manual Action:** Give acetaminophen
  - **Manual Action:** Give ibuprofen

KG ASSEMBLY RESOURCES

- SNOMED
  - Snowstorm: https://github.com/IHTSDO/snowstorm
  - BioPortal: https://data.bioontology.org/documentation
- Manually

LOGIC

- **1. KG-backed observation predicates:** These come from:
    - mapped caregiver language
    - KG normalization
    - thresholded criteria
  - Examples:
    - fever(child)
    - vomiting(child)
    - repeated_vomiting(child)
    - poor_intake(child)
    - no_urine_8h(child)
    - …
  - These are the **atoms** your rules operate over.
- **2. Intermediate clinical concern predicates:** These are useful because you keep the final rules clean.
  - Examples:
    - dehydration_concern(child)
    - respiratory_red_flag(child)
    - serious_ill_appearance(child)
  - These can also be defined in Datalog from the lower-level predicates.
- **3. Final disposition predicates**
  - These are your action categories.
  - Examples:
    - home_monitor(child)
    - call_doctor(child)
    - urgent_eval(child)
    - er_now(child)

EXAMPLE

- **From the CPG:**
  - “Is not alert when awake (lethargic)”
  - “Has not wet a diaper or gone pee in 8 hours”
  - “Has dry lips, tongue or mouth”
  - “Vomits often”
  - “Has trouble breathing”
  - “Has a fever of over 104 degrees F”
- **KG predicates**

  lethargy(C)
  trouble_breathing(C)
  no_urine_8h(C)
  repeated_vomiting(C)
  high_fever(C)
- **Intermediate rules**

  danger_red_flag(C) <= lethargy(C)
  danger_red_flag(C) <= trouble_breathing(C)
  danger_red_flag(C) <= high_fever(C)

  dehydration_concern(C) <= no_urine_8h(C)
  dehydration_concern(C) <= repeated_vomiting(C)
- **Final escalation rules**

  urgent_eval(C) <= dehydration_concern(C)
  er_now(C) <= danger_red_flag(C)

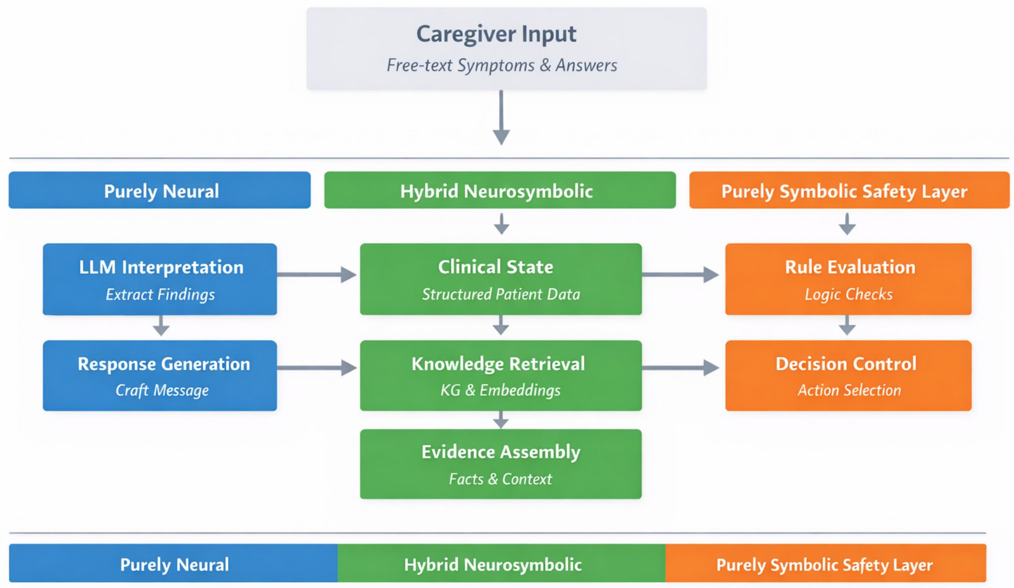
LAYERS OF RULES

- **Layer 1: observation / criterion predicates**
  - Derived from:
    - state
    - thresholds
    - KG mapping
  - Example:
    - high_fever
    - poor_intake
    - no_urine_8h
- **Layer 2: concern predicates**
  - Example:
    - dehydration_concern
    - danger_red_flag
- **Layer 3: disposition predicates**
  - Example:
    - home_monitor
    - urgent_eval
    - er_now
- **Layer 4: action predicates**
  - Example:
    - recommend_fluids
    - recommend_acetaminophen
    - recommend_er_now

SAFE & TRUSTWORTHY SYSTEM

- **Desired attributes**
  - **Confidence / Trustworthiness:** consistent, stable, clinically sensible plan
  - **Actionability:** clear steps + what to monitor + when to escalate
  - **Safety & Transparency:** explicit, unambiguous escalation triggers
- **Confidence / Trustworthiness**
  - grounded in clinical knowledge (SNOMED concepts + guidelines)
  - consistent reasoning across turns (no changing advice)
- **Actionability**
  - clear steps: what to do now
  - explicit: what to monitor + when to escalate
- **Safety & Transparency**
  - explicit red flags (not vague warnings)
  - clear “why” behind escalation decisions

ARCHITECTURE



- **Purely Neural (LLM layer)**
  - Interprets language → extracts candidate findings
  - Generates the **final caregiver-facing response**

- **Hybrid Neurosymbolic**
  - **Clinical State** → structured patient state (findings, unknowns, risk indicators)
  - **Knowledge Retrieval (KG + embeddings)** → pulls relevant concepts (red flags, discriminators, treatments)
  - **Evidence Assembly** → builds a grounded evidence set for reasoning
    - This is where: grounding happens , consistency is enforced, relevant knowledge is selected

- **Purely Symbolic (Safety layer)**
  - **Rule Evaluation (Datalog)** → applies escalation logic (red flags, thresholds)
  - **Decision Control** → final disposition: home / urgent / ER
  - This is where:
    - safety is enforced
    - decisions are authorized (not generated)

EXPLORE

- **Language → Concept Mapping**
  - Exact matching vs embedding-based mapping
  - Conservative mapping vs broader interpretation
  - Tradeoff: precision vs recall of findings
- **Knowledge Retrieval Strategy**
  - Minimal retrieval (only red flags + key questions)
  - Slightly richer retrieval (conditions, context)
  - Tradeoff: focus vs completeness
- **State Design Granularity**
  - Very minimal state (5–6 variables)
  - Slightly richer state (time, severity, intake, etc.)
  - Tradeoff: simplicity vs decision accuracy

EXPLORE

- **4. Rule Coverage & Strictness**
  - Conservative (escalate early)
  - Balanced (use more discriminators before escalation)
  - Tradeoff: safety vs over-escalation
- **5. Embeddings Usage (Targeted)**
  - Only for concept normalization
  - Also for question selection
  - Tradeoff: control vs flexibility
- **Response Framing (LLM layer)**
  - Minimal, directive responses
  - Slightly richer explanation with rationale
  - Tradeoff: clarity vs verbosity

“STATE”

- **Intuition**
  - “State” = **what the system currently knows about the child**
  - It is a **structured snapshot**, not conversation text
- **What goes into state**
  - **Symptoms / findings**
    - fever, vomiting, lethargy, breathing issues
  - **Key facts**
    - temperature range, duration, fluid intake, urination
  - **Red flags (if present)**
    - not alert, no urine, trouble breathing
  - **Unknowns (important!)**
    - breathing status?, urination?, alertness?
- **Why we need state**
  - Keeps the system **consistent across turns**
  - Helps decide **what to ask next**
  - Enables **rule-based decisions (safe, not guesswork)**
- **What state is NOT**
  - neither raw conversation text nor long descriptions
  - only **decision-relevant facts**
- *Léo Jacqmin, Lina M. Rojas Barahona, and Benoit Favre. 2022. “Do you follow me?”: A Survey of Recent Approaches in Dialogue State Tracking. In Proceedings of the 23rd Annual Meeting of the Special Interest Group on Discourse and Dialogue, pages 336–350, Edinburgh, UK. Association for Computational Linguistics.*

- **Example:** After 2 turns, state might look like:

```
fever = yes
vomiting = once
alert = yes
drinking = some
urination = unknown
```

- This is what drives:
  - next question
  - rule evaluation
  - final decision

AGENTIC WITH LANGGRAPH

- Why LangGraph for this project
  - LangGraph is built for stateful agents / workflows, which fits CareTrace's turn-by-turn triage loop.
- Your graph should be very simple:
  - START → interpret/update state → retrieve KG evidence → evaluate rules / choose next step → generate response → END or next turn
- In LangGraph terms, CareTrace mainly needs:
  - a StateGraph with a typed shared state,
  - a few nodes,
  - mostly normal edges plus one or two conditional edges for routing,
  - and short-term memory / checkpointing so the conversation state persists across turns.
- What "agentic" means here: A state-based control loop where each node does one job
  - update structured clinical state,
  - retrieve relevant knowledge,
  - apply rule logic,
  - decide whether to ask, reassure, or escalate,
  - then draft the caregiver-facing reply.

LANGGRAPH

- **Python LangGraph Overview:** Read this only for the big picture
- **Graph API Overview** (focus on)
    - **StateGraph**
    - **nodes**
    - **normal edges**
    - **conditional edges**
    - **START / END**
- **Most important implementation page**
- **Use the Graph API**
  This is the main page you should work from. It covers:
    - state
    - sequences
    - branches
    - loops
    - and the **Command API** for state update + routing.
- For CareTrace, you mainly need **state, branches, loops**, and possibly **Command** only if you want cleaner control flow.

LANGGRAPH

- For the basic build pattern
  - Quickstart

    Copy the pattern, not the exact agent:
    - define state
    - define nodes
    - define a routing function
    - build with StateGraph
    - compile the graph.
- For multi-turn conversation memory
  - Add Memory

    Read only the parts on:
    - short-term memory
    - thread-level persistence
    - checkpointer / InMemorySaver
    - (That is enough for this project; you do not need long-term memory)
- Links
  - https://docs.langchain.com/oss/python/langgraph/overview
  - https://docs.langchain.com/oss/python/langgraph/graph-api
  - https://docs.langchain.com/ (https://docs.langchain.com/oss/python/learn)

GROUPS OF FOUR MEMBERS EACH

- KG & Knowledge Lead
- Hybrid Reasoning Lead (LLM + KG + embeddings)
- Verification & Safety Lead
- Evaluation & System Integration Lead

EXTRA INVESTIGATION: PICK ONE

- **Option A. Reasoning Models (Neural Layer)**
  - We are investigating if stronger reasoning models improve:
    - selection of the *right questions*
    - interpretation of ambiguous caregiver input
  - Why : better neural reasoning should improve state quality, but must still respect symbolic decision rules
- **Option B. Logic-Augmented Generation (LAG)**
  - We are investigating if providing:
    - state + evidence + rules + decision
      to the LLM leads to:
    - more trustworthy and consistent explanations
  - Why : tests whether symbolic structure → better neural generation, rather than free-form responses
- **Option C. Ablation Study (System-Level Insight)**
  - We are investigating if each component actually contributes:
    - pure LLM
    - KG + LLM
    - KG + rules (full system)
  - Why: isolates the value of knowledge grounding and explicit rules in improving safety and trust